LTE MOBILITY

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FOR INDONESIAN TELECOMMUNICATION PROFESSIONALS (ITP)
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- WCDMA to LTE (W2L) Cell Reselection
- LTE to WCDMA (L2W) Cell Reselection
- Intra/Inter Frequency Cell Reselection
- Intra/Inter Frequency Handover via X2 and S1
- IRAT Handover
- Circuit Switch Fall Back (CSFB)
- Handover Parameter Optimization
- Handover KPIs
- Log File Analysis

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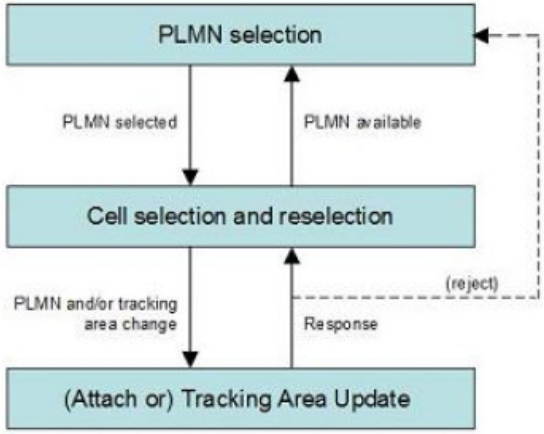
Idle Mode

- The idle mode algorithm ensures that the UE is camped on the best cell in idle mode from where it can reach and be reached by the network.



- The UE behavior in idle mode includes the following tasks:
  - PLMN selection
  - Cell selection and reselection
  - Tracking Area (TA) update

- The eNodeB broadcasts idle mode information that assists and controls the UE to select PLMN and best suitable LTE cell:

MIB Physical Layer Information	**SIB Type 1** Access restrictions, SIB scheduling information	**SIB Type 2** Common and shared channel information
SIB Type 3 Cell re-selection information	**SIB Type 4** Cell re-selection information, intra-frequency neighbor information	**SIB Type 5** Cell re-selection information, intra-frequency neighbor information
SIB Type 6 Cell re-selection information for UTRA	**SIB Type 7** Cell re-selection information for GERAN	**SIB Type 8** Cell re-selection information for CDMA2000
SIB Type 9 Home eNB identifier (HNBID)	**SIB Type 10** ETWS[1] primary notification	**SIB Type 11** ETWS[1] secondary notification

[1] ETWS = Earthquake and Tsunami Warning System

Cell Selection Criteria

The cell selection criteria is defined in 3GPP standards 36.304 UE Procedures in Idle Mode. It is based on the measured Reference Signal Received Power (RSRP) level in the cell.

The cell selection criterion is fulfilled if:
Srxlev > 0
where
Srxlev = Qrxlevmeas - (Qrxlevmin + Qrxlevminoffset) - Pcompensation

The UE obtains the Srxlev value using the following measurements and parameter values:
Qrxlevmeas: Measured RSRP value in the cell UE is evaluating (dBm)
Qrxlevmin: Required minimum RSRP level in the cell (dBm), indicated by cell in SIB1 message.
Qrxlevminoffset: Offset to Qrxlevmin taken into account in the Srxlev evaluation as a result of a periodic search for a higher priority PLMN
Pcompensation: Compensation [max (PEMAX - PUMAX, 0)] if the maximum power according to the UE capability (PUMAX) is less than the maximum UE power to be used in the cell (PEMAX)

LTE Cell Reselection Parameters

The UE is in idle mode in LTE. It starts measuring:

a) LTE neighbors when:

RSRP of best cell < Qrxlevmin (SIB3)+ Sintrasearch

Example: Qrxlevmin = -120 dBm

Sintrasearch = 62 dB

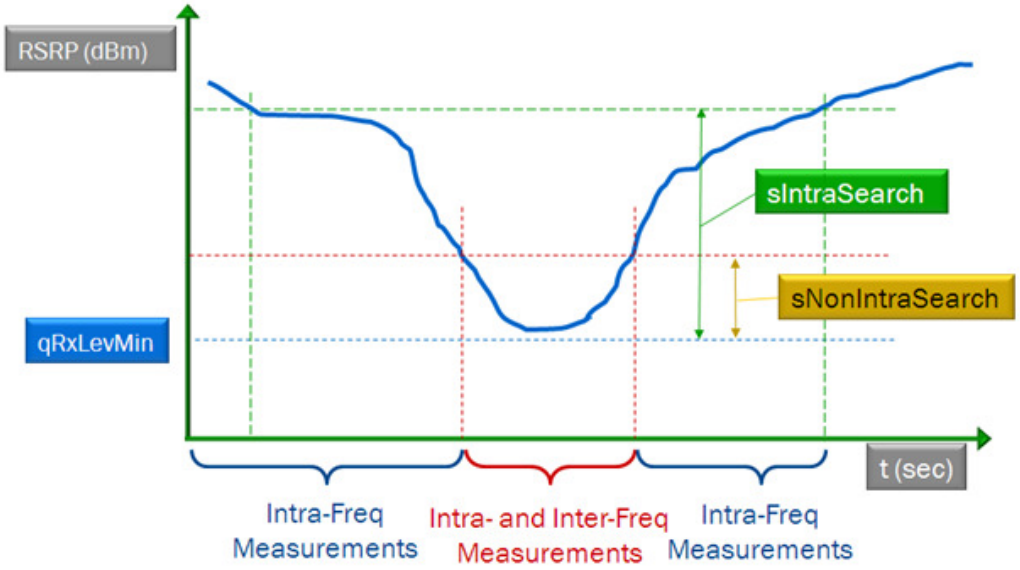
Then, the UE starts measuring RSRP of neighboring cells in LTE in the same frequency band.

b) IRAT neighbors when:

RSRP of best cell < Qrxlevmin (SIB3)+ SNonintrasearch

Example: Qrxlevmin (SIB3)= -120 dBm

Sintrasearch = 14 dB



qRxLevmin (in SIB3) defines the minimum RSRP values measured by the UE in a cell to be able to get unrestricted coverage-based service in that cell.
sIntraSearch, when added to the qRxLevmin value, will set the threshold for the UE to decide if it has to do intra-frequency cell measurements for potential cell reselection. If the current measured RSRP value for the cell is greater than the threshold set up by the sIntraSearch parameter, then the UE is not required to do intra-frequency measurements. If the current value of RSRP measured by the UE drops below the line, then the UE is required to do intra-frequency measurements for potential cell reselection.
sNonIntraSearch is similar in nature to sIntraSearch except that it dictates when the UE has to do inter-frequency measurements for cell reselection purposes. Note that, when inter-frequency measurements are done, the UE will still continue to do intra-frequency measurements as well.
Threshservinglow when added to qrxlevmin indicates the best cell reception level (RSRP) used in reselection evaluation towards 1xEV-DO. That is, if the RSRP measured is below this threshold, the UE is in conditions to reselect to 1xEV-DO if other conditions are also met (See below). The value sent over the RRC interface is half the value configured (the UE then multiplies the received value by 2)
ThreshXLow represents the minimum level the Ec/Io of the 1xEV-DO pilot must have so that the UE decides to reselect 1xEV-DO rather than LTE.
Treselection is the time that the RSRP of the best cell must be under Threshservinglow and the measured Ec/Io of the 1xEV-DO pilot must be above ThreshXLowHRPD so that the UE decides to reselect 1xEV-DO.

LTE Intra-frequency Cell Reselection

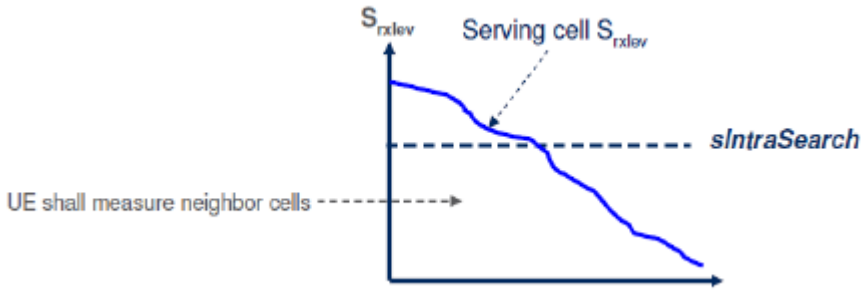
- Triggering of intra-frequency neighbor cell measurement

$S_{ServingCell} = Srxlev$

$S_{ServingCell}$ <= ***SIntraSearch***: **perform measurements**

$S_{ServingCell}$ > ***SIntraSearch***: **no measurements**

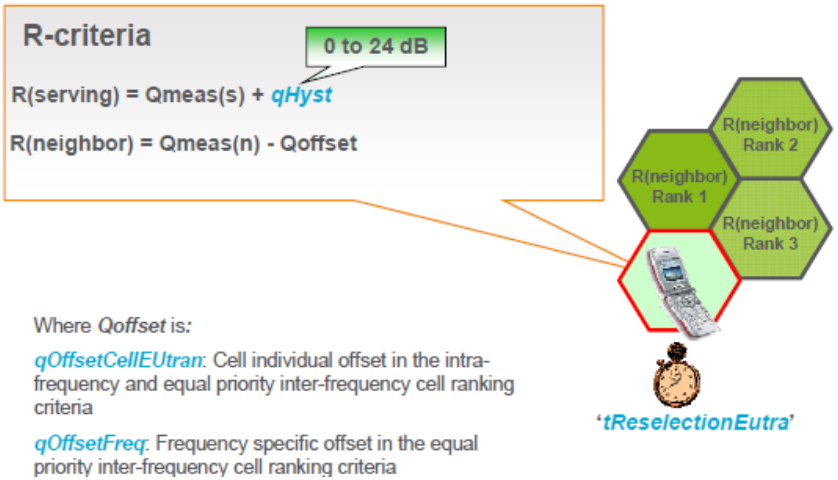
where,
sIntraSearch [62dB] – the threshold for intra-frequency measurements (*3GPP TS 36.304*



- To minimize the impact on UE battery performance, it is possible to set the threshold above which UE may not measure intra-frequency neighbor cells.
- When the UE triggers a cell reselection evaluation process, it ranks cells that fulfill the cell selection criteria. The UE ranks the cells according to the Rcriteria.

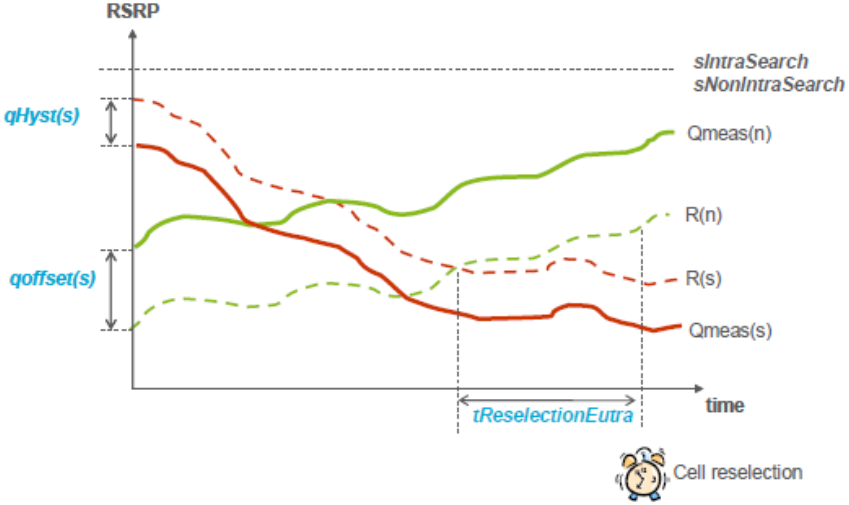
Cell Reselection Evaluation Process

- The UE performs ranking of all cells fulfilling the S-criteria (S > 0). The UE ranks the cells according to the 'R-criterion'.



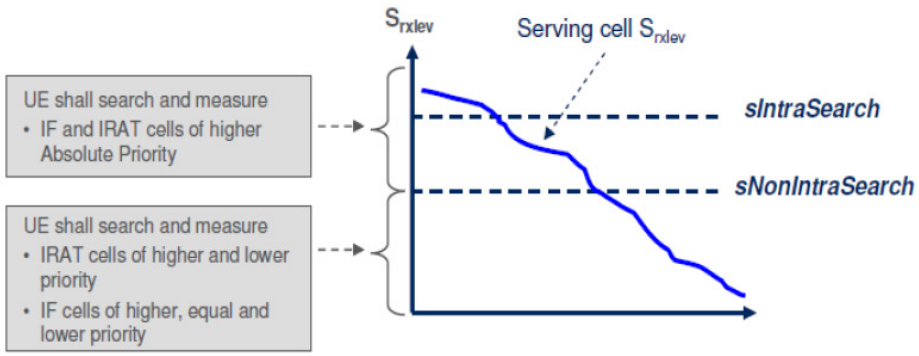
- **Qmeas** is the quality value of the received signal. (RSRP)
- **qHyst(s)** is the hysteresis value (qHyst) that is read in the system information of the serving cell.
- **Qoffset** is an offset in the cell ranking criterion of neighbor EUTRAcells. It consists of a cell individual part and a frequency specific part. The frequency specific part applies to equal priority inter-frequency cells only.

- The Qhyst is a hysteresis value preventing too frequent reselection back and forth between cells of nearly equal rank. When a neighboring cell is ranked as better than the serving cell (that is, Rn > Rs) during a time interval tReselectionEutra, the UE performs a cell reselection to the better ranked cell.



LTE to WCDMA Cell Reselection

- Triggering of Inter-frequency and IRAT cell search and measurements



Srxlev = Qrxlevmeas – **qRxLevMin** [EUtranCellFDD]

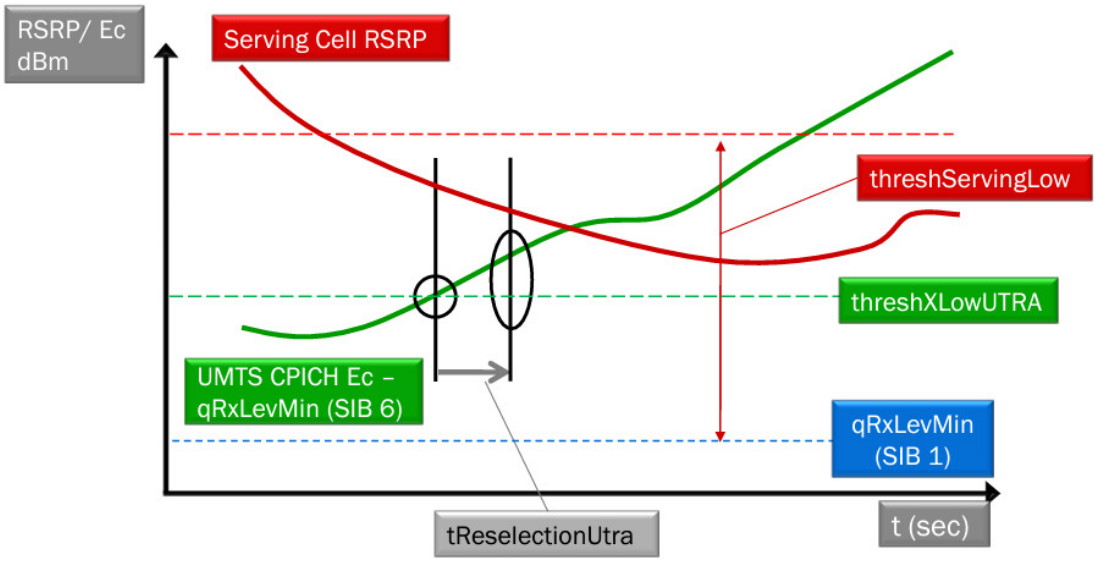
- A WCDMA-capable Rel-8 UE in LTE shall always perform measurements on WCDMA frequencies with higher priority than the serving LTE cell as defined in SIB6.
- The intensity of these measurements may vary depending on whether Srxlev of LTE-serving cell is above or below the thresholds defined by the**sNonIntraSeach** (RSRP threshold) parameter. For higher priority WCDMAfrequencies, **sNonIntraSearch** determines below which connection quality level the measurements should be intensified.When it comes to lower priority WCDMA frequencies, the same parameter determines if the UE needs to measure at all. Measurements are mandatory only when Srxlev is below **sNonIntraSearch**.

Cell Reselection to Lower Priority WCDMA Frequency

Inter-RAT cell reselection to a lower priority WCDMA frequency is performed by the UE in LTE when the LTE and WCDMA criteria listed below are fulfilled during **tReselectionUtra** seconds:

- Serving LTE cell RSRP below **qRxLevMin** [EUtranCellFDD] + **threshServingLow**
- Target WCDMA cell RSCP above **qRxLevMin** [UtranFreqRelation] +**threshXLow**

If more than one WCDMA cell meets the cell reselection criteria above, the UE shall reselect to the WCDMA cell with the highest measured RSCP level, or more precise highest Srxlev. See *Idle Mode and Common Channel Behavior* for the definition of Srxlev (and Squal) for serving WCDMA cell.



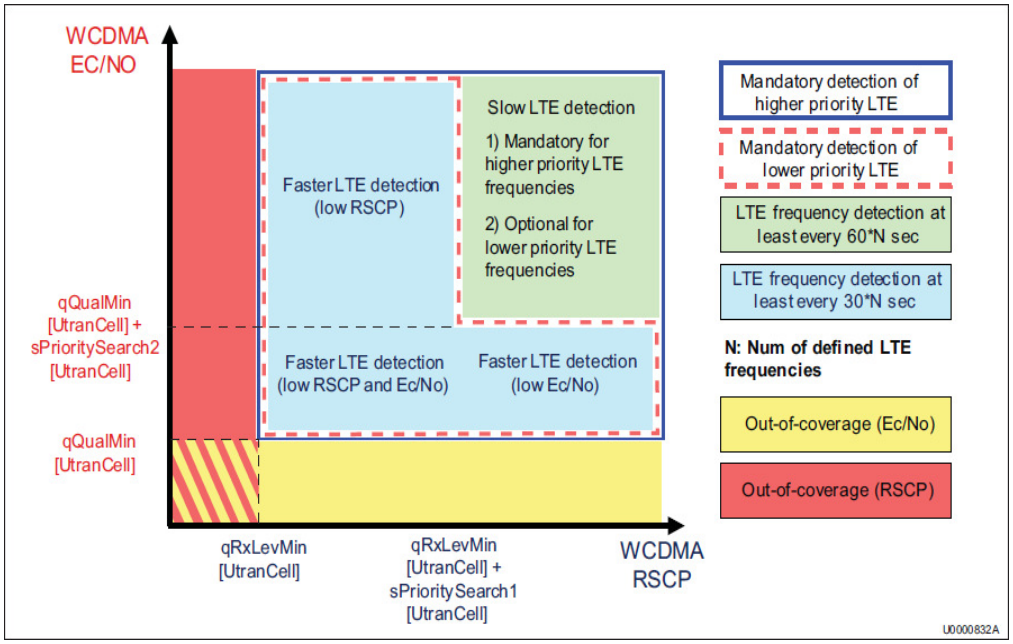
Priority-Based Cell Reselection – WCDMA to LTE

UE must measure the LTE frequencies and detect the available LTE cell in order to perform cell reselection to LTE.

UE measures two physical properties called for WCDMA signal. One is CPICH RSCP and CPICH EcNo. RSCP determines Srxlev and EcNo determines Squal.

Srxlev = Qrxlevemeas - qRxLevMin. Qrxlevemeas is RSCP level measured by UE and qRxLevMin is the value specified in SIB.
Squal = Qqualmeas - qQualMin. Qqualmeas is EcNo level measured by UE and qQualMin is the value specified in SIB.

The detection measurement of LTE frequencies should be done at least once every 60s for higher priority LTE frequencies.



In following condition, detection measurements of lower priority LTE frequency is not required.
Srxlev > absPrioCellRes.sPrioritySearch1
Squal > absPrioCellRes.sPrioritySearch2

In following condition, UE should detect once every 30s for both lower and higher priority LTE frequencies
Srxlev <= absPrioCellRes.sPrioritySearch1
Squal <= absPrioCellRes.sPrioritySearch2

The maximum number of LTE FDD Frequencies are 4. In this case, UE should have performed measurement for detecting LTE cells on all 4 LTE frequencies once every 240 (4 x 60) s or 120(4 x 30) s depending if UE measures above or below parameter threshold absPrioCellRes.sPrioritySearch1

Measuring LTE Cell

Once the UE has detected an LTE cell, it is required to periodically perform LTE cell measurements. The measurement time interval is determined based on configured DRX cycle length values.

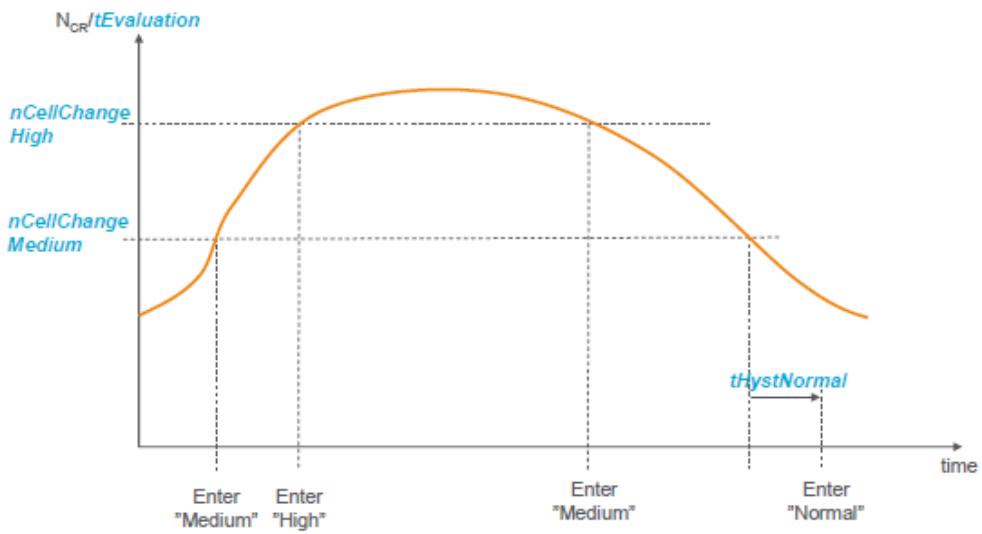
The DRX cycle length value used for idle mode W2L cell reselection is the lowest value comparing the setting of **cnDrxCycleLengthCs** and **cnDrxCycleLengthPs**. The measurement time interval used for W2L cell reselection in state URA_PCH is determined by **utranDrxCycleLength**. With current WRAN recommended DRX settings, measurement time interval, TmeasureE-UTRA, is 5.12 s for both idle mode and URA_PCH.

DRX cycle length [s]	$T_{measureE\text{-}UTRA}$ [s] (number of DRX cycles)
0.08	2.56 (32)
0.16	2.56 (16)
0.32	5.12 (16)
0.64	5.12 (8)
1.28	6.4 (5)
2.56	7.68 (3)
5.12	10.24 (2)

Speed-Dependent Cell Reselection

- The speed-dependent scaling of cell reselection criteria is used to influence the cell reselection criteria for fast moving UE. It helps the UE to respond more quickly to cell changes when moving at high speed. A UE may enter three different mobility states:
  - Normal mobility
  - Medium mobility
  - High mobility

- The Medium and High states are specified by the parameter *nCellChangeMedium* and *nCellChangeHigh* resepectively defining number of cell reselections within sliding time window *tEvalution* [240sec] that determines the UE shall enter mobility states Medium/High.



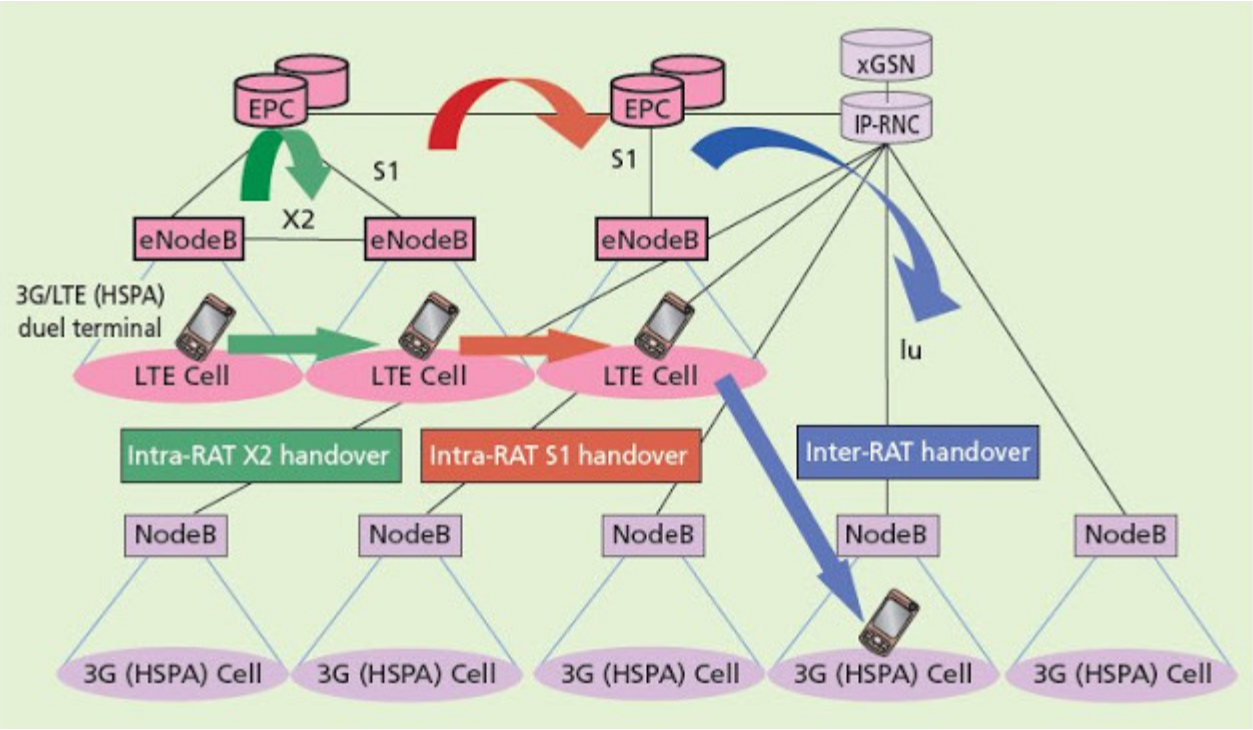
Overview LTE Handovers

1) ***Intra-RAT handover***: This is a handover that occurs within the LTE system in which node transition occurs between sectors within an eNodeB, between eNodeBs within an EPC switch, or between EPC switches.

A handover between eNodeBs within an EPC switch may be an X2 or S1 handover. In an X2 handover, signal processing is performed by the X2 logical interface between eNodeBs, while in an S1 handover, signal processing is performed by the S1 logical interface between an eNodeB and the EPC switch. There is a tradeoff between the cost of maintaining an X2 link and the cost incurred by an S1 handover, and operations are configured accordingly.

Handover can also be classified by whether the center frequency is the same before and after handover, that is, whether the handover occurs within the same frequency or between frequencies.

2) ***Inter-RAT handover***: This is a handover that occurs between RATs either as a transition from LTE to 3G or from 3G to LTE.

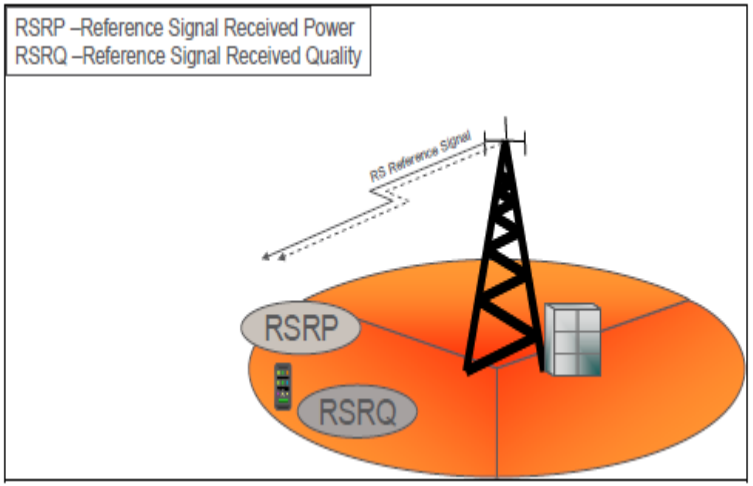


LTE Measurements

- The Intra-LTE Handover feature is based on measurements and coverage triggers evaluated by the user equipment. The user equipment measurements are reported to the serving RBS which makes the ultimate decision on inter-cell handover.

- Two types of measurements are used in the handover evaluation process:
  - **RSRP** (Reference Signal Received Power) which represents the mean measured power per reference symbol
  - **RSRQ** (Reference Signal Received Quality) which provides an indication of the reference signal quality

- The Intra-LTE handover can be set to trigger on the *RSRP* value or the RSRQ value and the measurement reports sent by the user equipment contain either or both of these values



RSRP is a RSSI type of measurement. It measures the average received power over the resource elements that carry cell-specific reference signals within certain frequency bandwidth. RSRP is applicable in both RRC_idle and RRC_connected modes, while RSRQ is only applicable in RRC_connected mode. In the procedure of cell selection and cell reselection in idle mode, RSRP is used.

RSRQ is a C/I type of measurement and it indicates the quality of the received reference signal. It is defined as (N*RSRP)/(E-UTRA Carrier RSSI), where N makes sure the nominator and denominator are measured over the same frequency bandwidth;

Measurement Report: LTE Events

0 PERIODICAL:
Periodical reporting.

1 **EVENT_A1**
Serving cell becomes better than absolute threshold.

2 **EVENT_A2**
Serving cell becomes worse than absolute threshold.

3 **EVENT_A3**
Neighbor cell becomes amount of offset better than serving.

4 **EVENT_A4**
Neighbor cell becomes better than absolute threshold.

5 **EVENT_A5**
Serving cell becomes worse than absolute threshold1 AND Neighbor cell becomes better than another absolute threshold2.

LTE Events IRAT

0 PERIODICAL
Periodical reporting.

1 **EVENT_B1**
IRAT neighbour becomes better than threshold

2 **EVENT_B2**
Serving becomes worse than threshold1 and IRAT neighbour becomes better than threshold2

Handover Stages

1. Setting up measurements
2. Evaluating measurement reports
3. Handover Preparation
4. Handover Execution
5. Handover Completion
6. Data forwarding

- If inter-RAT is to GSM, NACC(Network assisted Cell Change) is used and no resources are prepared in the target system. UE enter GSM system in RRC_Idle mode and start the RRC Connection Setup Procedure.

Intra LTE Handover

3GPP TS 36.331

- The Intra-LTE Handover feature configures EventA3 as defined in *3GPP TS 36.331*. The EventA3 implies that one or several neighbor cells become better than the serving cell also when some offset and hysteresis values are taken into account – "neighbor becomes offset better than serving".

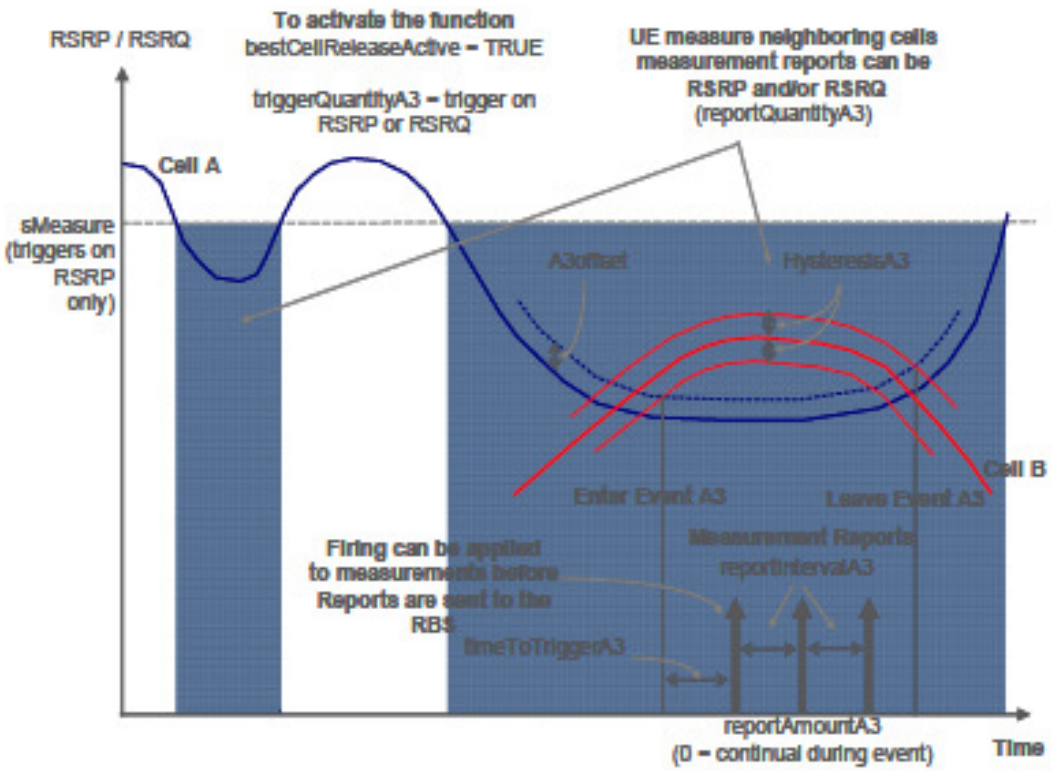
- The process employed by the user equipment for the evaluation of surrounding cells uses parameters sent by the serving RBS to the user equipment. These parameters include hysteresis and offset values, time to trigger, and optionally cell individual offset margins.

- The Intra-LTE Handover feature is based on the evaluations reported to the RBS by the user equipment. The serving RBS uses the reports to select and prepare the target RBS, then ultimately conducts conclusion of the handover.

- User equipment use two alternative types of measurements in the cell evaluation process:
  - Reference Signal Received Power (RSRP) representing the mean measured power per reference signal
  - Reference Signal Received Quality (RSRQ) providing an indication ofthe reference signal quality

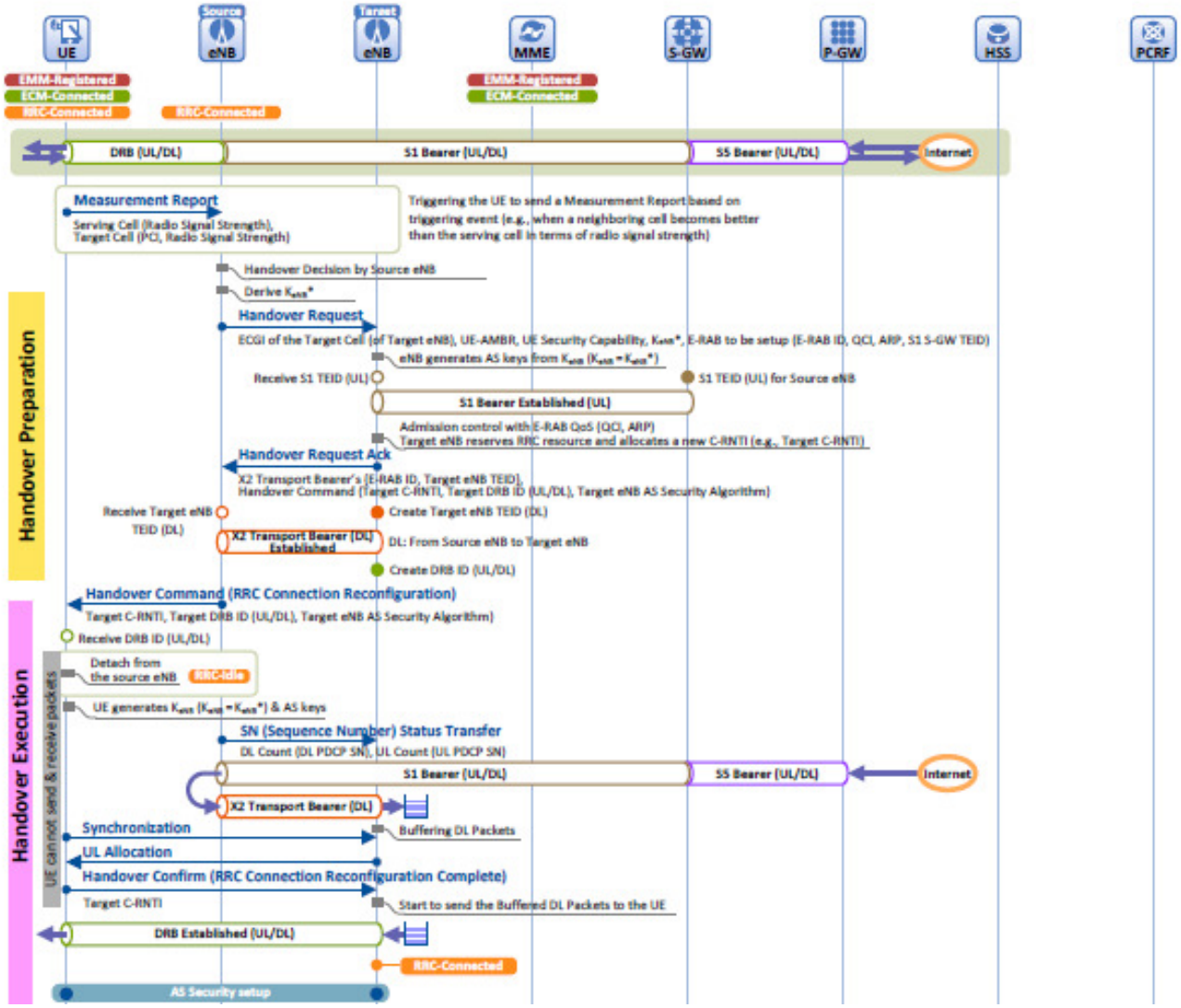
Event A3

Event A3 is defined as a triggering event when a neighbor cell becomes better than the serving cell.



$$Mn + cellIndividualOffsetEUtran - hysteresisA3 > Ms + a3Offset$$

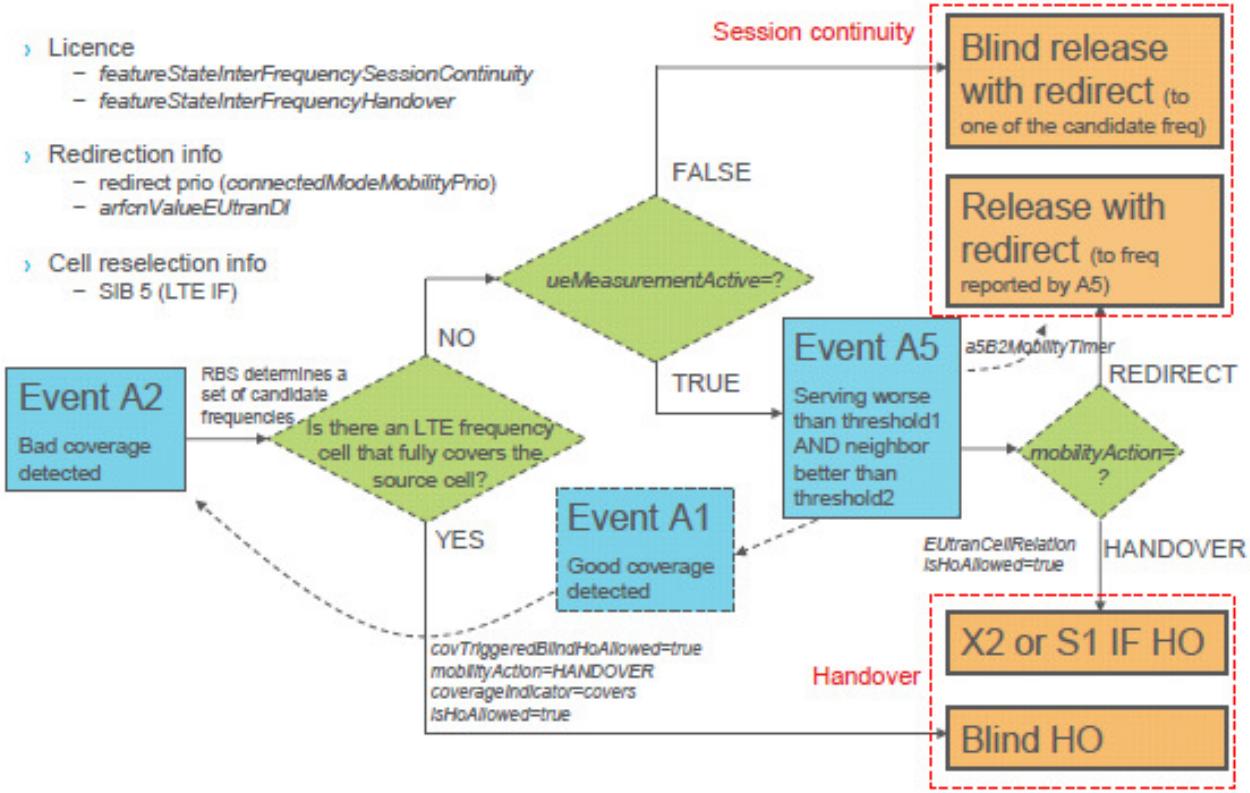
LTE X2 Handover Call Flow



Inter Frequency Handover

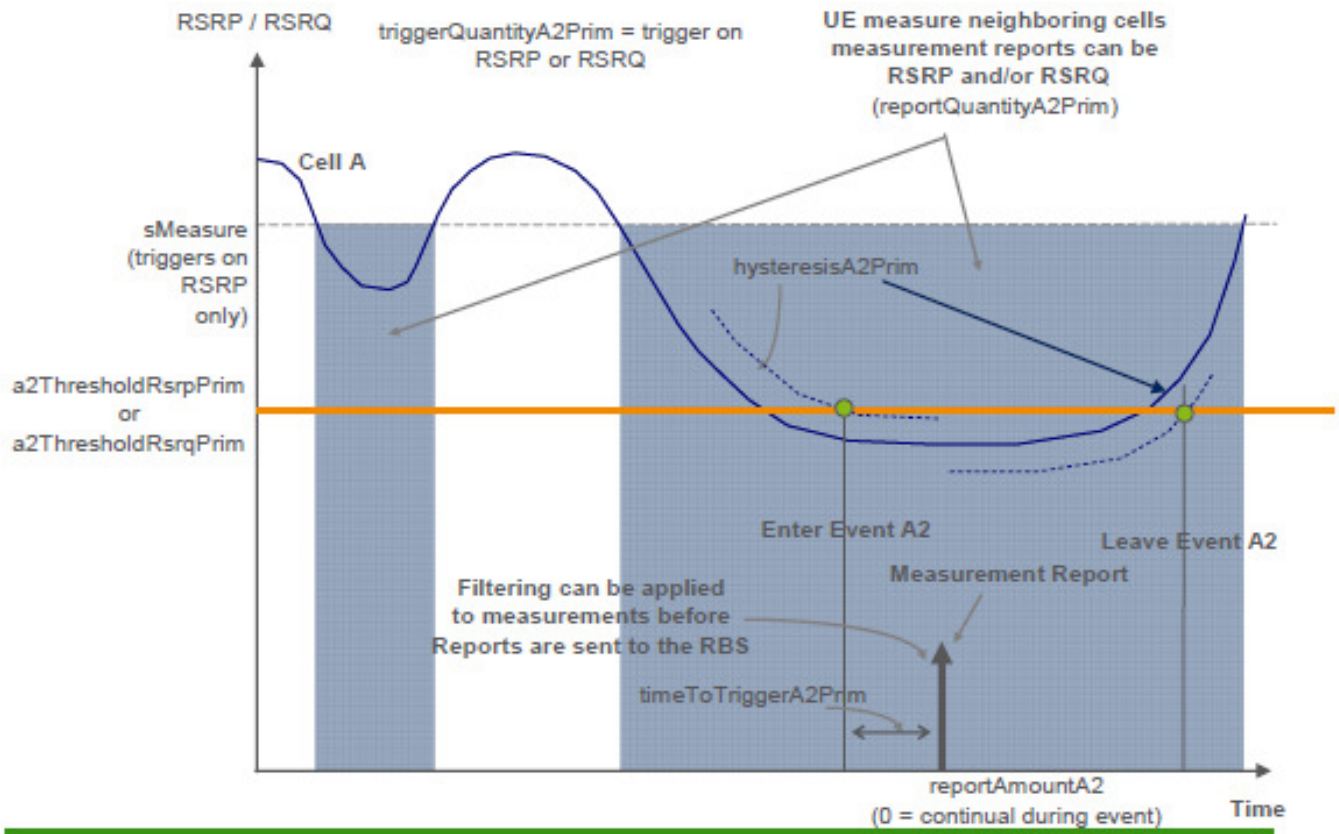
- The purpose of Inter-Frequency Handover is to extend the call with the option of initiating a handover to a cell on a different LTE frequency, instead of initiating a release.

- The Inter Frequency Mobility consists of
  - Coverage Triggered Inter Frequency Handover
  - Coverage Triggered Inter Frequency Session Continuity



Event A2

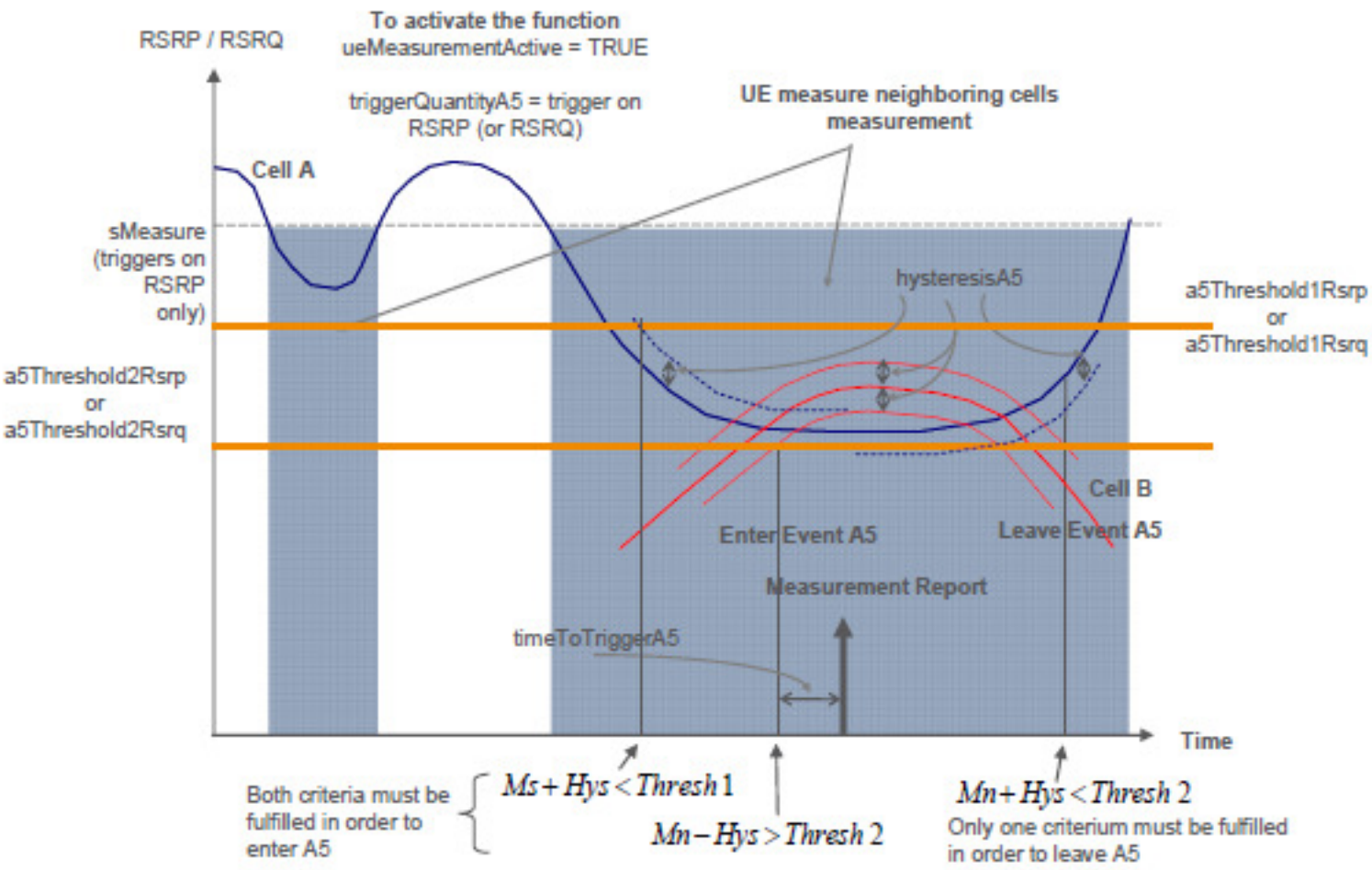
Event A2 defined as a triggering event when the serving cell drops below a certain threshold.



$$Ms + hysteresisA2Prim < a2ThresholdPrimary$$

Event A5

Event A5 is used by Inter Frequency Handover in order to find a frequency to redirect to or a cell to make handover to.



Handover KPIs

Ericsson based counters.

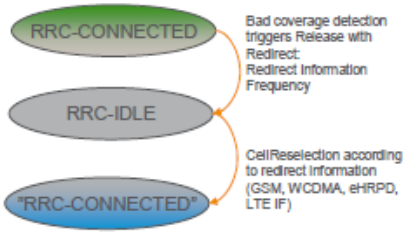
KPI_GROUP	KPI_NAME	FORMULA/COUNTER	Unit	DESCRIPTION
Mobility	% INTRAF_HO_SUCC_RATE	100*((PMHOPREPSUCCLTEINTRAF/PMHOPREPATTLTEINTRAF)*(PMHOEXESUCCLTEINTRAF/PMHOEXEATTLTEINTRAF))	%	Intra frequency handover success rate
Mobility	% HO_PREP_INTRAF_SUCC_RATE	100*(PMHOPREPSUCCLTEINTRAF/PMHOPREPATTLTEINTRAF)	%	Handover preparation intra frequency success rate
Mobility	% HO_EXEC_INTRAF_SUCC_RATE	100*(PMHOEXESUCCLTEINTRAF/PMHOEXEATTLTEINTRAF)	%	Handover execution intra frequency success rate
Mobility	% INTERF_HO_SUCC_RATE	100*((PMHOPREPSUCCLTEINTERF/PMHOPREPATTLTEINTERF)*(PMHOEXESUCCLTEINTERF/PMHOEXEATTLTEINTERF))	%	Inter frequency handover success rate
Mobility	% HO_PREP_INTERF_SUCC_RATE	100*(PMHOPREPSUCCLTEINTERF/PMHOPREPATTLTEINTERF)	%	Handover preparation inter frequency success rate
Mobility	% HO_EXEC_INTERF_SUCC_RATE	100*(PMHOEXESUCCLTEINTERF/PMHOEXEATTLTEINTERF)	%	Handover execution inter frequency success rate
Mobility	HoPrepRejInLicMob	pmHoPrepRejInLicMob	#	The number of Incoming Handover Preparation Failure messages sent by the target eNB due to lack of license for handover.
Mobility	HoPrepRejInLicConnUsers	pmHoPrepRejInLicConnUsers	#	The number of Handover Preparation Failure messages sent by the target eNB due to license for connected users being exceeded.
Mobility	HoPrepRejInLicRlcUm	pmHoPrepRejInLicRlcUm	#	The number of non-admitted E-RABs at an incoming handover preparation procedure, due to lack of license for RLC UM.
Mobility	HoPrepRejInLicMultiErab	pmHoPrepRejInLicMultiErab	#	The number of non-admitted E-RABs at an incoming handover preparation procedure, due to license for Multiple E-RAB being exceeded.

IRAT Handover

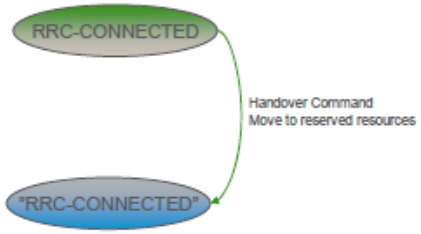
There are in principle two ways of inter-working between LTE and other RATs.

1) The inter-working can be performed by a prepared handover (network controlled) where the UE does not leave the CONNECTED state (Handover)
2) By a cell reselection (UE controlled or network assisted) where UE via IDLE state performs network assisted cell reselection.

› Release with Redirect (Coverage Triggered Session Continuity)



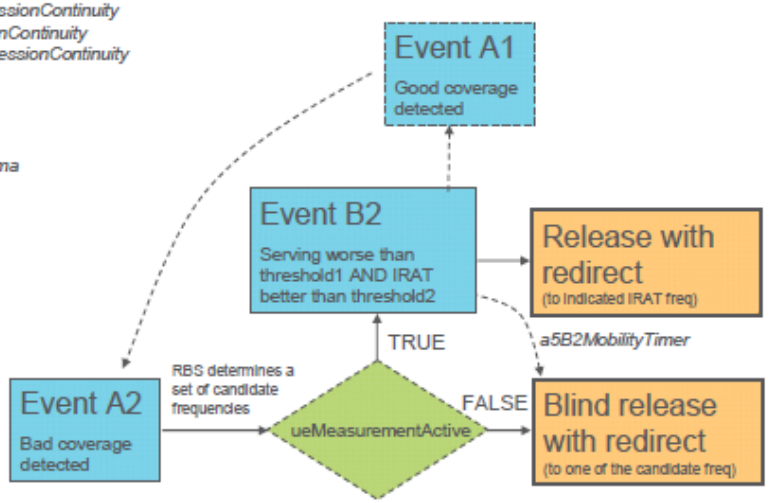
› Handover



› Licence
- *featureStateWCDMASessionContinuity*
- *featureStateGsmSessionContinuity*
- *featureStateCDMA200sessionContinuity*

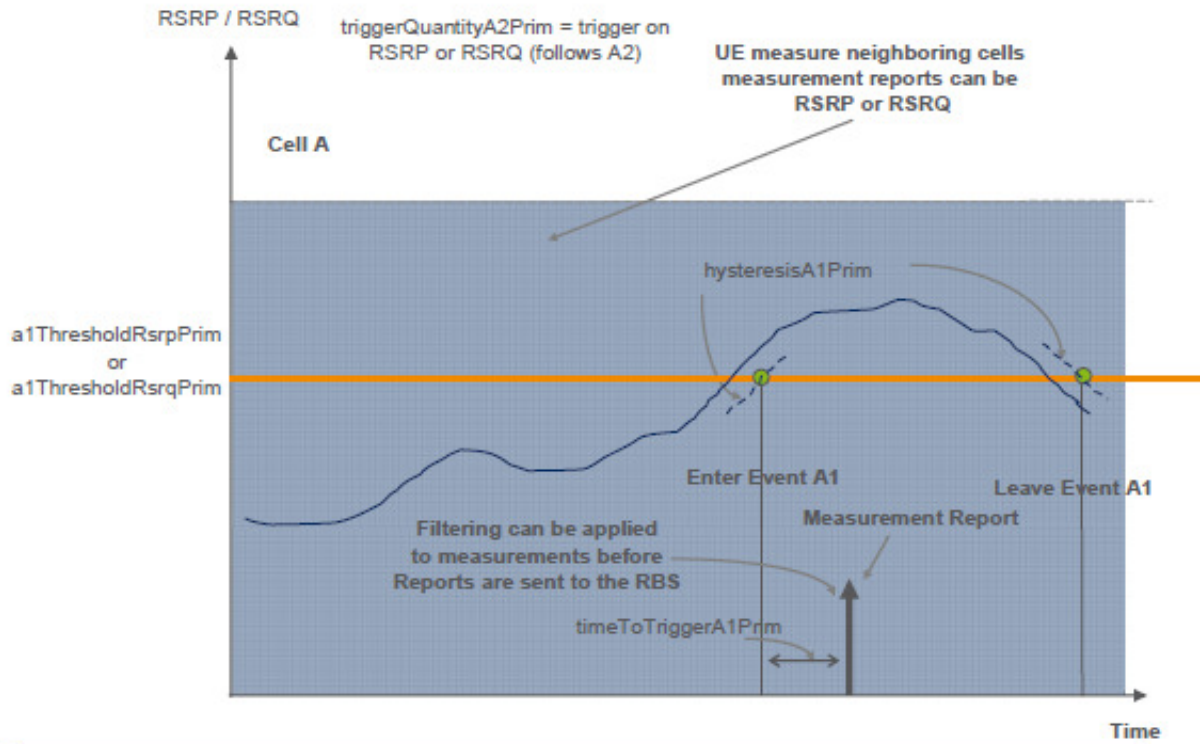
› Redirection info
- *arfcnValueUtranDl*
- *arfcnValueGeranDl*
- *hrpdBandClass, freqCdma*

› Cell reselection info
- SIB 6 (WCDMA)
- SIB 7 (GERAN)
- SIB 8 (CDMA)



Event A1

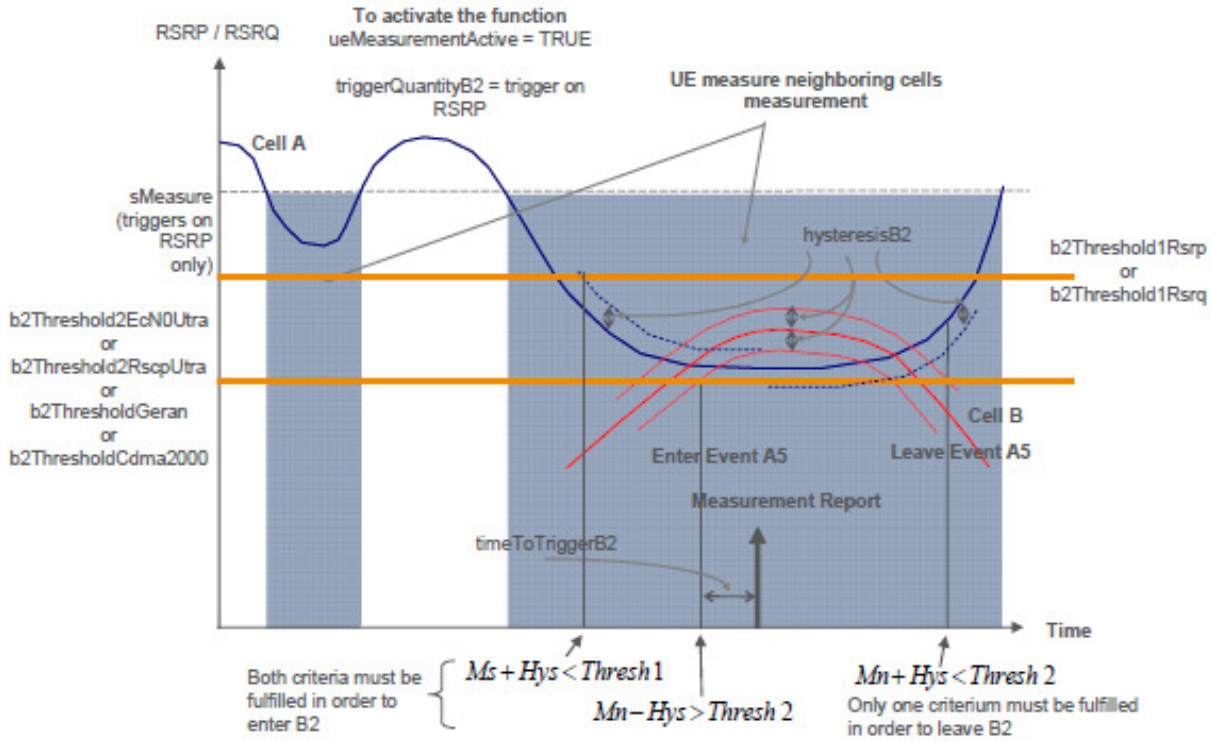
The redirection to another frequency may be cancelled by event A1 (“Good Coverage”).



Ms-hysteresisA2Prim < a2ThresholdPrimary

Event B2

Event B2 is used by IRAT Coverage Triggered Session Continuity in order to decide which type of release (blind or not) will be performed.



IRAT KPIs

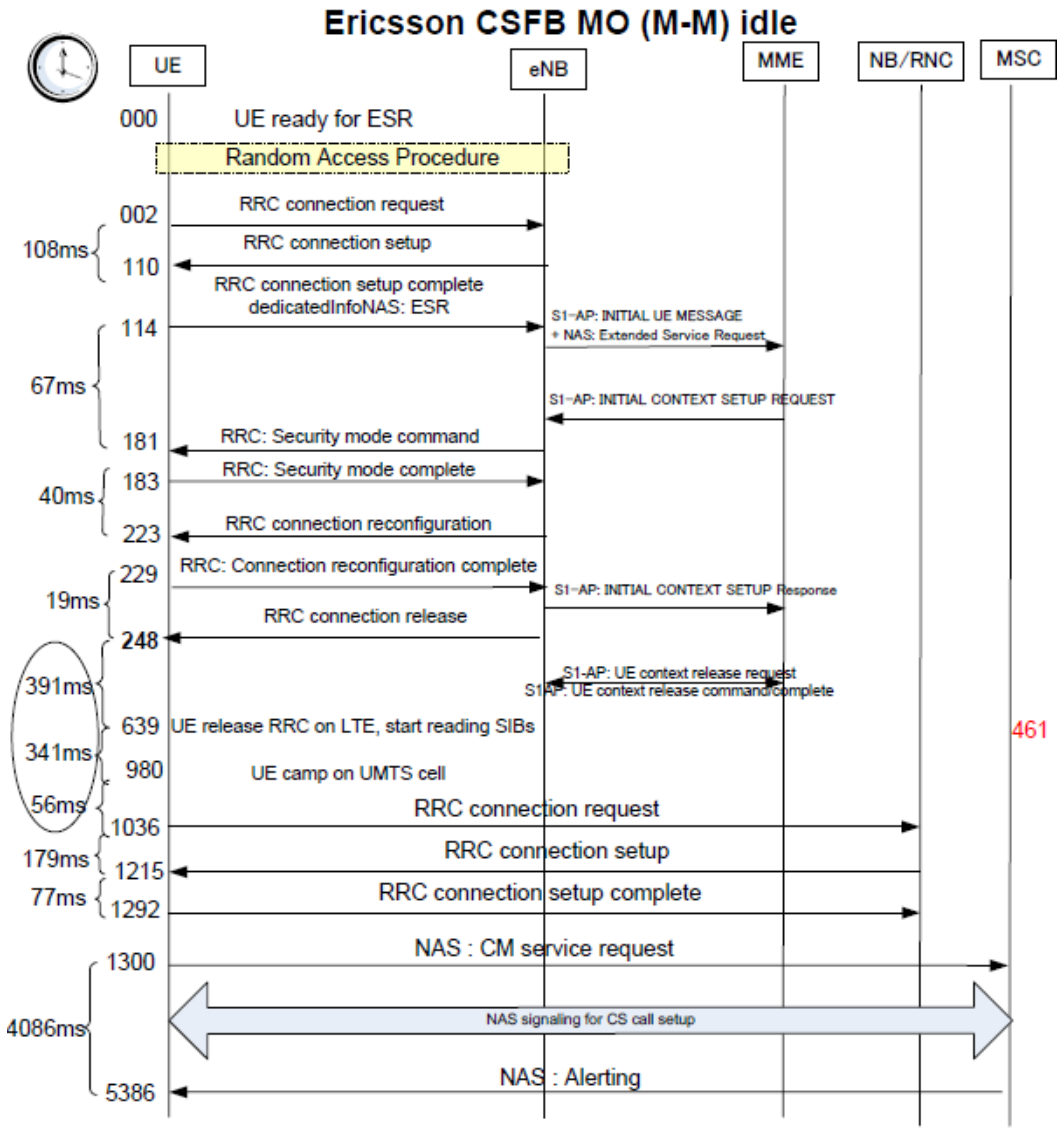
KPI_GROUP	KPI_NAME	FORMULA/COUNTER	Unit	DESCRIPTION
Mobility	% IRAT_Redirect	100*(pmUeCtxtRelSCGsm + pmUeCtxtRelSCWcdma)/(pmUeCtxtEstabSucc)	#	
Mobility	BadCovEvalReport	pmBadCovEvalReport		The total number of traffic Bad Coverage measurement reports received. Condition: Stepped at reception of traffic measurement report from UE indicating that bad coverage due to Event A2, 'Serving becomes worse than absolute threshold'.

Circuit Switch Fallback (CSFB)

3GPP TS 23.272

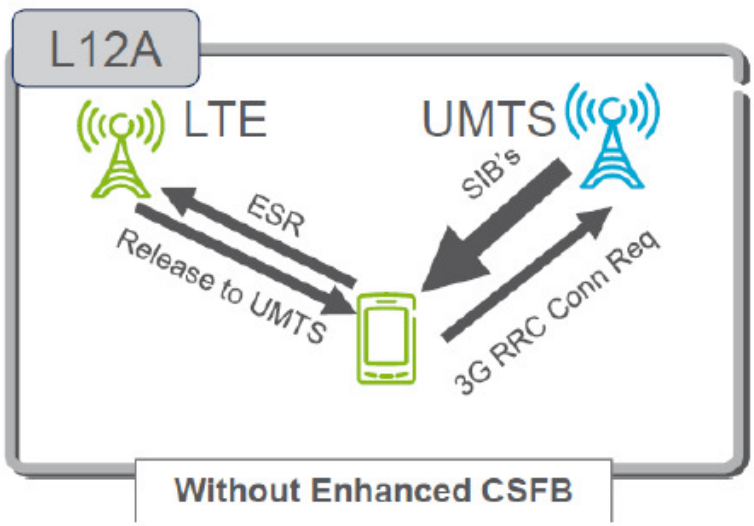
- The purpose of the CS Fallback to GSM and WCDMA is to enable traditional CS-domain voice services for LTE devices.
- CS fallback is the only method to provide traditional voice services for single radio UE unless IP Multimedia Subsystem (IMS) Voice over IP (VoIP) services are available.
- The CS Fallback function is of course only possible to realize in areas where E-UTRAN coverage is overlapped with GSM and WCDMA coverage.
- The benefit of CS Fallback to GSM or WCDMA is that it is possible to offer voice services in E-UTRAN without requiring IMS VoIP support.
- The CS Fallback to GSM or WCDMA feature provides a basic mobility between an LTE cell and neighboring GSM or WCDMA cells. When this feature is active an LTE terminal will be released from the LTE network and attach to the neighboring GSM or WCDMA network with minimal disruption for the end user.

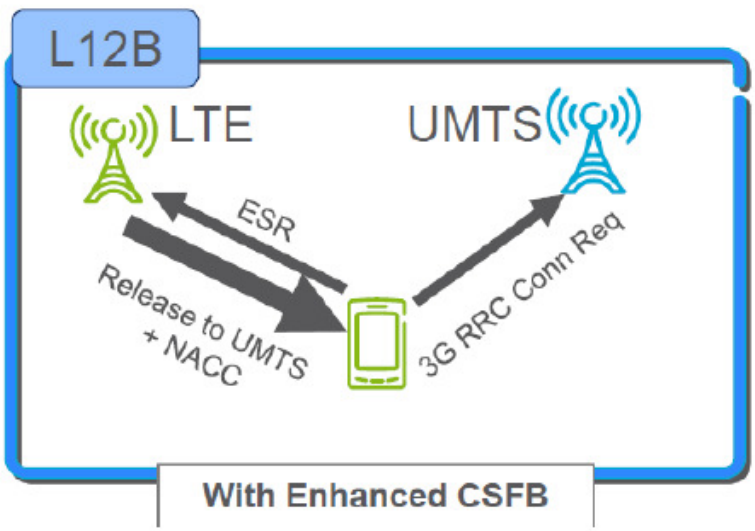
CSFB Call Flow



Enhanced Circuit Switch Fall Back (eCSFB)

- eCSFB uses RAN Information Method (RIM) procedure to get SIB information from RNC in UMTS network
- SIB information is sent to the UE on LTE throught "RRC Connection Release" message
- The goal is to reduce call setup time by avoiding the UE to read SIB over the air on UMTS
- Working for CSFB LTE to UMTS only
- According to study call set up can be reduced by 400ms – 500ms





CSFB KPIs

Ericsson based counters.

KPI_GROUP	KPI_NAME	FORMULA/COUNTER	Unit	DESCRIPTION
Mobility	% CSFB_NORM_ERAB_SUCC	100*((pmUeCtxtRelCsfbWcdma+pmUeCtxtRelCsfbWcdmaEm+ PMUECTXTRELCSFBGSM+PMUECTXTRELCSFBGSMEM)/ pmUeCtxtEstabSucc)	%	The percentage CSFB normailzed to UE context established success.
Mobility	CSFB_GSM_ATTS	PMUECTXTRELCSFBGSM	#	The number of RRC Connection Release with redirect to GSM triggered by CS Fallback.
Mobility	CSFB_GSM_EMERG_ATTS	PMUECTXTRELCSFBGSMEM	#	The number of RRC Connection Release with redirect to GSM triggered by CS Fallback Emergency.
Mobility	CSFB_WCDMA_ATTS	pmUeCtxtRelCsfbWcdma	#	The number of RRC Connection Release with redirect to WCDMA triggered by CS Fallback.
Mobility	CSFB_WCDMA_EMERG_ATTS	pmUeCtxtRelCsfbWcdmaEm	#	The number of RRC Connection Release with redirect to WCDMA triggered by CS Fallback Emergency.

Thank you.

Question and Answer